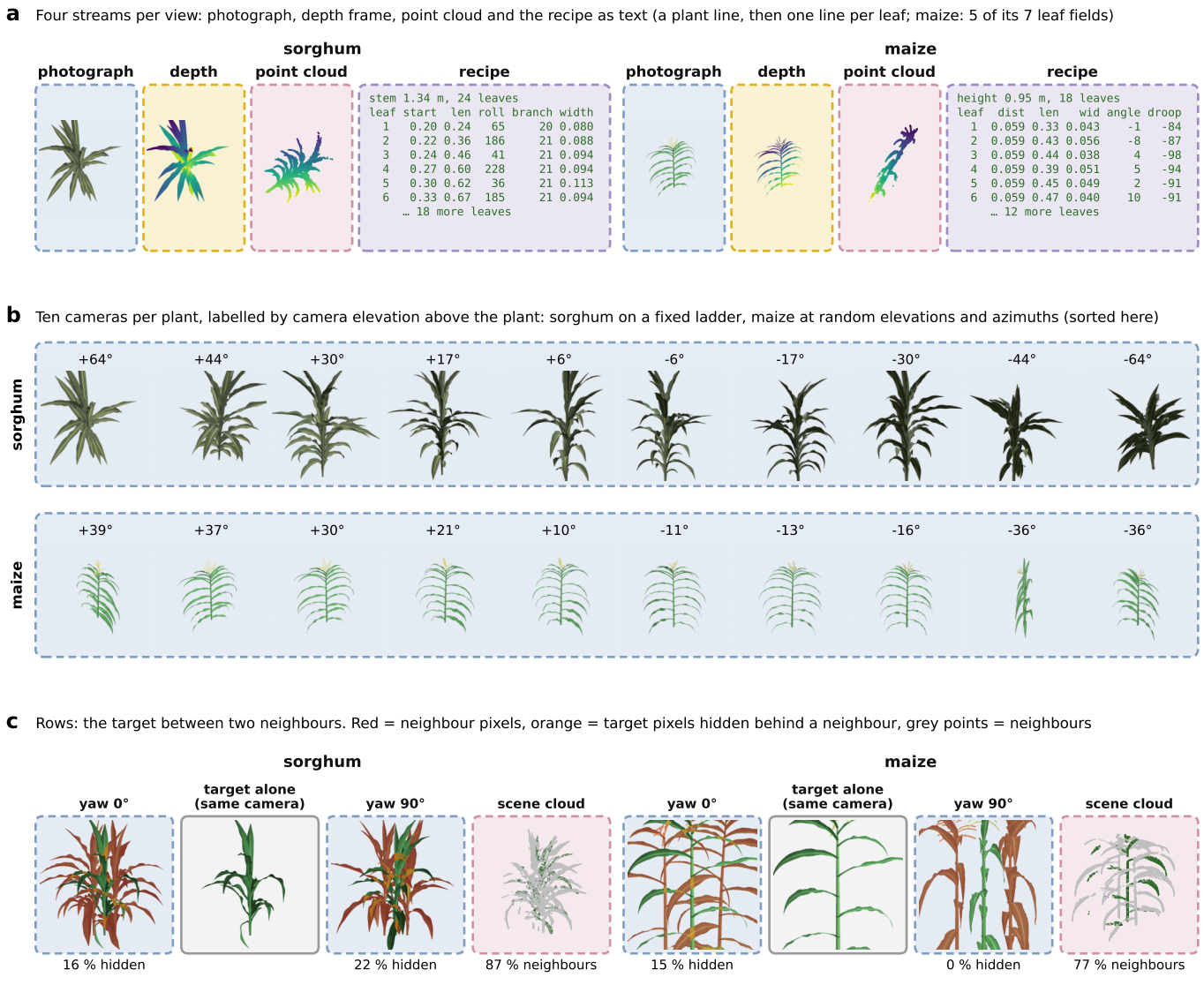

a
Four streams per view: photograph, depth frame, point cloud and the recipe as text (a plant line, then one line per leaf; maize: 5 of its 7 leaf fields)
sorghum
photograph
depth
point cloud
recipe
stem 1.34 m, 24 leaves
leaf start len roll branch width
1 0.20 0.24 65 20 0.080
2 0.22 0.36 186 21 0.088
3 0.24 0.46 41 21 0.094
4 0.27 0.60 228 21 0.094
5 0.30 0.62 36 21 0.113
6 0.33 0.67 185 21 0.094
… 18 more leaves
maize
photograph
depth
point cloud
recipe
height 0.95 m, 18 leaves
leaf dist len wid angle droop
1 0.059 0.33 0.043 -1 -84
2 0.059 0.43 0.056 -8 -87
3 0.059 0.44 0.038 4 -98
4 0.059 0.39 0.051 5 -94
5 0.059 0.45 0.049 2 -91
6 0.059 0.47 0.040 10 -91
… 12 more leaves
b
Ten cameras per plant, labelled by camera elevation above the plant: sorghum on a fixed ladder, maize at random elevations and azimuths (sorted here)
sorghum
+64° +44° +30° +17° +6° -6° -17° -30° -44° -64°
maize
+39° +37° +30° +21° +10° -11° -13° -16° -36° -36°
c
Rows: the target between two neighbours. Red = neighbour pixels, orange = target pixels hidden behind a neighbour, grey points = neighbours
sorghum
yaw 0°
target alone (same camera)
yaw 90°
scene cloud
16 % hidden
22 % hidden
87 % neighbours
maize
yaw 0°
target alone (same camera)
yaw 90°
scene cloud
15 % hidden
0 % hidden
77 % neighbours